BTS(C) – IV– 10 – 012 – G

B. Tech Degree IV Semester Examination, April 2010

CE 402 A/B SURVEYING II
(2006 Scheme)

Time: 3 Hours | Maximum Marks: 100

PART A
(Answer ***ALL*** questions)

(8 x 5 = 40)

I. a. Explain the principle and procedure for setting out a simple curve by two theodolite method.
b. Enumerate various types of vertical curves with sketches.
c. List out the factors to be considered for the selection of triangulation figures.
d. Form the normal equations for a, b, c and d in the following equations

Equation	Weight
$3a + 4b + 3c - d - 5 = 0$	1
$a - 2b + 4c + 2d - 1 = 0$	2
$2a + b + 2c + 5d - 2 = 0$	3

e. Differentiate between solar apparent time and mean solar time.
f. Explain the following terms:
(i) Equation of time (ii) Sidereal day (iii) Equinoctial points
(iv) Zenith distance (v) Solastices
g. Describe the location of sounding stations by means of
(i) Goss rope soundings (ii) Intersecting ranges
h. Write a note on the various branches of terrestrial photogrammetry.

PART B

(4 x 15 = 60)

II. Two roads meet at an angle of $127^0 30'$. Calculate the necessary data for setting out a curve of 100m radius to connect the two straight portions of the road if a theodolite is available. (15)

OR

III. Explain three methods for determining length of a transition curve. (15)

IV. a. List out the factors to be considered for the selection of triangulation stations. (7)
b. Two triangulation stations A and B are 60 kilometers apart and have elevations 240m and 280m respectively. Find the minimum height of signal required at B so that the line of sight may not pass near the ground than 2 meters. The intervening ground may be assumed to have a uniform elevation of 200 meters. (8)

OR

V. a. Explain the procedure to determine the most probable value of observations using method of differences. (4)
b. The following angles were measured at a station 0 so as to close the horizon:

$\angle AOB = 83^0\ 42'\ 28.75''$ weight 3

$\angle BOC = 102^0\ 15'\ 43.26''$ weight 2

$\angle COD = 94^0\ 38'\ 27.22''$ weight 4

$\angle DOA = 79^0\ 23'\ 23.77''$ weight 2

Adjust the angles (11)

(Turn over)

VI. a. Discuss the various co-ordinate systems used for specifying the position of a celestial body. (8)

b. The attitudes of a star at upper and lower transits of a star are $70^0\ 20'$ and $20^0\ 40'$, both the transits being on the north side of zenith of the place. Find the declination of the star and the latitude of the place of observation. (7)

OR

VII. a. List out the properties of a spherical triangle. (5)

b. Find the zenith distance and altitude at upper culmination of the stars from the following data:

(i)	Declination of star = $42^0\ 15'$ N	Latitude of observer = $26^0\ 40'$ N
(ii)	Declination of star = $23^0\ 20'$ N	Latitude of observer = $26^0\ 40'$ N
(iii)	Declination of star = $65^0\ 40'$ N	Latitude of observer = $26^0\ 40'$ N

(10)

VIII. Differentiate between air photograph and a map. (15)

OR

IX. Explain any five methods to determine the levels of points on the river bed and fix the position of the soundings. (15)

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